Chapter 5

Spatial Electron Distributions

Some of this is listed in tables in Foot [2]. Please contact Prof. Campbell if you find any errors.

5.1 Wavefunctions

The wavefunctions are most-commonly expressed as complex eigenstates of L_z or as real-valued functions that point along directions in space. The complex ones are typically preferred in physics, and are the energy eigenstates of gas-phase atoms, with or without a static, uniform magnetic field. The real-valued functions are typically preferred in chemistry, where symmetry breaking from chemical bonding makes the energy eignstates more-closely resemble the real-valued functions.

First, we can write some of the lowest-energy complex wavefunctions. The general case is given by

$$\begin{aligned}\psi_{n,L,M}(\mathbf{r}) &= \langle \mathbf{r}|\psi_{n,L,M}\rangle \\ &= \sqrt{\left(\frac{2Z}{na_0}\right)^{3+2L} \frac{(n-L-1)!}{(n+L)!\,2n}}\; e^{-Zr/(na_0)}\; r^L\, \mathscr{L}_{n-L-1}^{2L+1}\left[\frac{2Z}{na_0} r\right]\; Y_{LM}(\boldsymbol{\omega})\end{aligned} \tag{5.1}$$

where $\mathscr{L}_n^a[x]$ is the generalized Laguerre polynomial in x (we are using the convention adopted by, for example, Mathematica).

orbital	$\langle \mathbf{r} \vert \psi_{n,L,M} \rangle$
$1s$ $(M_L = 0)$	$\sqrt{\frac{Z^3}{\pi a_0^3}}\ e^{-Zr/a_0}$
$2s$ $(M_L = 0)$	$\frac{1}{4}\sqrt{\frac{Z^5}{2\pi a_0^5}}\ \left(r - 2\frac{a_0}{Z}\right)\ e^{-Zr/(2a_0)}$
$2p$ $(M_L = \pm 1)$	$\mp\frac{1}{8}\sqrt{\frac{Z^5}{\pi a_0^5}}\ r\ e^{-Zr/(2a_0)}\ \sin(\theta)\, e^{\pm i\phi}$
$2p$ $(M_L = 0)$	$\frac{1}{4}\sqrt{\frac{Z^5}{2\pi a_0^5}}\ r\ e^{-Zr/(2a_0)}\ \cos(\theta)$
$3s$ $(M_L = 0)$	$\frac{2}{81}\sqrt{\frac{Z^7}{3\pi a_0^7}}\left(r^2 - 9\frac{a_0}{Z}\, r + \frac{27}{2}\left(\frac{a_0}{Z}\right)^2\right) e^{-Zr/(3a_0)}$
$3p$ $(M_L = \pm 1)$	$\mp\frac{1}{81}\sqrt{\frac{Z^7}{\pi a_0^7}}\left(r - 6\frac{a_0}{Z}\right)\ r\, e^{-Zr/(3a_0)}\ \sin(\theta)\, e^{\pm i\phi}$
$3p$ $(M_L = 0)$	$\frac{1}{81}\sqrt{\frac{2Z^7}{\pi a_0^7}}\left(r - 6\frac{a_0}{Z}\right)\ r\, e^{-Zr/(3a_0)}\ \cos(\theta)$
$3d$ $(M_L = \pm 2)$	$\frac{1}{162}\sqrt{\frac{Z^7}{\pi a_0^7}}\ r^2\, e^{-Zr/(3a_0)}\ \sin^2(\theta)\, e^{\pm i2\phi}$
$3d$ $(M_L = \pm 1)$	$\mp\frac{1}{81}\sqrt{\frac{Z^7}{\pi a_0^7}}\ r^2\, e^{-Zr/(3a_0)}\ \sin(\theta)\cos(\theta)\, e^{\pm i\phi}$
$3d$ $(M_L = 0)$	$\frac{1}{81}\sqrt{\frac{Z^7}{6\pi a_0^7}}\ r^2\, e^{-Zr/(3a_0)}\ (3\cos^2(\theta) - 1)$
$4s$ $(M_L = 0)$	$\frac{1}{1536}\sqrt{\frac{Z^9}{\pi a_0^9}}\ \left(r^3 - 24\frac{a_0}{Z}\, r^2 + 144\left(\frac{a_0}{Z}\right)^2\ r - 192\left(\frac{a_0}{Z}\right)^3\right)\ e^{-Zr/(4a_0)}$
$4p$ $(M_L = \pm 1)$	$\mp\frac{1}{512}\sqrt{\frac{Z^9}{10\pi a_0^9}}\ \left(r^2 - 20\frac{a_0}{Z}\, r + 80\left(\frac{a_0}{Z}\right)^2\right)\ r\, e^{-Zr/(4a_0)}\ \sin(\theta)\, e^{\pm i\phi}$
$4p$ $(M_L = 0)$	$\frac{1}{512}\sqrt{\frac{Z^9}{5\pi a_0^9}}\ \left(r^2 - 20\frac{a_0}{Z}\, r + 80\left(\frac{a_0}{Z}\right)^2\right)\ r\, e^{-Zr/(4a_0)}\cos(\theta)$
$4d$ $(M_L = \pm 2)$	$\frac{1}{1024}\sqrt{\frac{Z^9}{6\pi a_0^9}}\ \left(r - 12\frac{a_0}{Z}\right)\ r^2\, e^{-Zr/(4a_0)}\ \sin^2(\theta)\, e^{\pm i2\phi}$
$4d$ $(M_L = \pm 1)$	$\mp\frac{1}{512}\sqrt{\frac{Z^9}{6\pi a_0^9}}\ \left(r - 12\frac{a_0}{Z}\right)\ r^2\, e^{-Zr/(4a_0)}\ \sin(\theta)\cos(\theta)\, e^{\pm i\phi}$

orbital	$\langle\mathbf{r}\vert\psi_{n,L,M}\rangle$
$4d$ ($M_L = 0$)	$\frac{1}{3072}\sqrt{\frac{Z^9}{\pi a_0^9}}\,\left(r - 12\frac{a_0}{Z}\right)\,r^2\,e^{-Zr/(4a_0)}\,\left(3\cos^2(\theta) - 1\right)$
$4f$ ($M_L = \pm 3$)	$\mp\,\frac{1}{6144}\sqrt{\frac{Z^9}{\pi a_0^9}}\,r^3\,e^{-Zr/(4a_0)}\,\sin^3(\theta)\,e^{\pm i3\phi}$
$4f$ ($M_L = \pm 2$)	$\frac{1}{1024}\sqrt{\frac{Z^9}{6\pi a_0^9}}\,r^3\,e^{-Zr/(4a_0)}\,\sin^2(\theta)\cos(\theta)\,e^{\pm i2\phi}$
$4f$ ($M_L = \pm 1$)	$\mp\,\frac{1}{2048}\sqrt{\frac{Z^9}{15\pi a_0^9}}\,r^3\,e^{-Zr/(4a_0)}\,\left(5\cos^2(\theta) - 1\right)\sin(\theta)\,e^{\pm i\phi}$
$4f$ ($M_L = 0$)	$\frac{1}{3072}\sqrt{\frac{Z^9}{5\pi a_0^9}}\,r^3\,e^{-Zr/(4a_0)}\,\left(5\cos^2(\theta) - 3\right)\cos(\theta)$

$R_{n,l}$	radial wavefunction
$R_{1,0}(r) =$	$2\sqrt{\frac{Z^3}{a_0^3}}\;e^{-Zr/a_0}$
$R_{2,0}(r) =$	$\frac{1}{2}\sqrt{\frac{Z^5}{2a_0^5}}\;\left(r - 2\frac{a_0}{Z}\right)\;e^{-Zr/(2a_0)}$
$R_{2,1}(r) =$	$\frac{1}{2}\sqrt{\frac{Z^5}{6a_0^5}}\;r\,e^{-Zr/(2a_0)}$
$R_{3,0}(r) =$	$\frac{4}{81}\sqrt{\frac{Z^7}{3a_0^7}}\;\left(r^2 - 9\frac{a_0}{Z}\,r + \frac{27}{2}\left(\frac{a_0}{Z}\right)^2\right)\;e^{-Zr/(3a_0)}$
$R_{3,1}(r) =$	$\frac{2}{81}\sqrt{\frac{2Z^7}{3a_0^7}}\;r\left(r - 6\frac{a_0}{Z}\right)\;e^{-Zr/(3a_0)}$
$R_{3,2}(r) =$	$\frac{2}{81}\sqrt{\frac{2Z^7}{15a_0^7}}\;r^2\,e^{-Zr/(3a_0)}$
$R_{4,0}(r) =$	$\frac{1}{768}\sqrt{\frac{Z^9}{a_0^9}}\;\left(r^3 - 24\frac{a_0}{Z}\,r^2 + 144\left(\frac{a_0}{Z}\right)^2 r - 192\left(\frac{a_0}{Z}\right)^3\right)\;e^{-Zr/(4a_0)}$
$R_{4,1}(r) =$	$\frac{1}{256}\sqrt{\frac{Z^9}{15a_0^9}}\;r\left(r^2 - 20\frac{a_0}{Z}\,r + 80\left(\frac{a_0}{Z}\right)^2\right)\;e^{-Zr/(4a_0)}$
$R_{4,2}(r) =$	$\frac{1}{768}\sqrt{\frac{Z^9}{5a_0^9}}\;r^2\left(r - 12\frac{a_0}{Z}\right)\;e^{-Zr/(4a_0)}$
$R_{4,3}(r) =$	$\frac{1}{768}\sqrt{\frac{Z^9}{35a_0^9}}\;r^3\,e^{-Zr/(4a_0)}$

The spherical harmonics, $Y_{L,M_L}(\boldsymbol{\omega})$, up to $L = 4$ (which is to say for s, p,

d, f, and g states) are given below.

$$\begin{aligned}
Y_{0,0}(\boldsymbol{\omega}) &= \frac{1}{2\sqrt{\pi}} \\
Y_{1,\pm1}(\boldsymbol{\omega}) &= \mp\frac{1}{2}\sqrt{\frac{3}{2\pi}}\;\sin(\theta)\,e^{\pm i\phi} \\
Y_{1,0}(\boldsymbol{\omega}) &= \frac{1}{2}\sqrt{\frac{3}{\pi}}\;\cos(\theta) \\
Y_{2,\pm2}(\boldsymbol{\omega}) &= \frac{1}{4}\sqrt{\frac{15}{2\pi}}\;\sin^2(\theta)\,e^{\pm i2\phi} \\
Y_{2,\pm1}(\boldsymbol{\omega}) &= \mp\frac{1}{2}\sqrt{\frac{15}{2\pi}}\;\cos(\theta)\sin(\theta)\,e^{\pm i\phi} \\
Y_{2,0}(\boldsymbol{\omega}) &= \frac{1}{4}\sqrt{\frac{5}{\pi}}\;\big(3\cos^2(\theta)-1\big) \\
Y_{3,\pm3}(\boldsymbol{\omega}) &= \mp\frac{1}{8}\sqrt{\frac{35}{\pi}}\;\sin^3(\theta)\,e^{\pm i3\phi} \\
Y_{3,\pm2}(\boldsymbol{\omega}) &= \frac{1}{4}\sqrt{\frac{105}{2\pi}}\;\cos(\theta)\sin^2(\theta)\,e^{\pm i2\phi} \\
Y_{3,\pm1}(\boldsymbol{\omega}) &= \mp\frac{1}{8}\sqrt{\frac{21}{\pi}}\;\big(5\cos^2(\theta)-1\big)\sin(\theta)\,e^{\pm i\phi} \\
Y_{3,0}(\boldsymbol{\omega}) &= \frac{1}{4}\sqrt{\frac{7}{\pi}}\;\big(5\cos^2(\theta)-3\big)\cos(\theta) \\
Y_{4,\pm4}(\boldsymbol{\omega}) &= \frac{3}{16}\sqrt{\frac{35}{2\pi}}\;\sin^4(\theta)\,e^{\pm i4\phi} \\
Y_{4,\pm3}(\boldsymbol{\omega}) &= \mp\frac{3}{8}\sqrt{\frac{35}{\pi}}\;\sin^3(\theta)\cos(\theta)\,e^{\pm i3\phi} \\
Y_{4,\pm2}(\boldsymbol{\omega}) &= \frac{3}{8}\sqrt{\frac{5}{2\pi}}\;\sin^2(\theta)\big(7\cos^2(\theta)-1\big)\;e^{\pm i2\phi} \\
Y_{4,\pm1}(\boldsymbol{\omega}) &= \mp\frac{3}{8}\sqrt{\frac{5}{\pi}}\;\sin(\theta)\big(7\cos^3(\theta)-3\cos(\theta)\big)\;e^{\pm i\phi} \\
Y_{4,0}(\boldsymbol{\omega}) &= \frac{3}{16\sqrt{\pi}}\;\big(35\cos^4(\theta)-30\cos^2(\theta)+3\big)
\end{aligned}$$

From the spherical harmonics, it is possible to define analogous functions that are real, and these end up being preferred in chemistry. The normalized tesseral harmonics are given by

$$Z^{\mathrm{c}}_{L,M}(\hat{\mathbf{e}}_r) \equiv \frac{1}{\sqrt{2}}\Big(Y_{L,-M}(\boldsymbol{\omega}) + (-1)^M Y_{L,M}(\boldsymbol{\omega})\Big) \tag{5.2}$$

$$Z_{L,0}(\hat{\mathbf{e}}_r) \equiv Y_{L,0}(\boldsymbol{\omega}) \tag{5.3}$$

$$Z^{\mathrm{s}}_{L,M}(\hat{\mathbf{e}}_r) \equiv \frac{i}{\sqrt{2}}\Big(Y_{L,-M}(\boldsymbol{\omega}) - (-1)^M Y_{L,M}(\boldsymbol{\omega})\Big) \tag{5.4}$$

where $M \geq 0$. Here, the "c" and "s" labels denote cosine and sine (of ϕ), respectively, and the middle line is to be used when $M = 0$. These are normalized to $\int \mathrm{d}^2\boldsymbol{\omega}\, Z^{t'}_{L',M'}\, Z^{t}_{L,M} = \delta_{L'L}\,\delta_{M'M}\,\delta_{t't}$ where $t, t' \in \{\mathrm{c}, \mathrm{s}\}$ (or nothing for $M = 0$).

The tesseral harmonics vanish on $L - M$ half-cones of rotation about the z axis (in a globe-surface model, $L - M$ parallels of latitude) and M planes that contain the z axis (M great circles of longitude). These planes intersect one another at locally-right angles on the surface of a sphere.

We can calculate a few of these to get a grasp of what spatial distributions are described by them. For this, we will make use of the Cartesian variables via substitution of

$$\begin{aligned} \frac{x}{r} &= \sin(\theta)\cos(\phi) \\ \frac{y}{r} &= \sin(\theta)\sin(\phi) \\ \frac{z}{r} &= \cos(\theta). \end{aligned} \tag{5.5}$$

These linear combinations give

$$
\begin{aligned}
Z_{0,0}(\hat{\mathbf{e}}_r) &= \frac{1}{2\sqrt{\pi}} \\
Z^{\mathrm{c}}_{1,1}(\hat{\mathbf{e}}_r) &= \frac{1}{2}\sqrt{\frac{3}{\pi}}\ \frac{x}{r} \\
Z^{\mathrm{s}}_{1,1}(\hat{\mathbf{e}}_r) &= \frac{1}{2}\sqrt{\frac{3}{\pi}}\ \frac{y}{r} \\
Z_{1,0}(\hat{\mathbf{e}}_r) &= \frac{1}{2}\sqrt{\frac{3}{\pi}}\ \frac{z}{r} \\
Z^{\mathrm{c}}_{2,2}(\hat{\mathbf{e}}_r) &= \frac{1}{4}\sqrt{\frac{15}{\pi}}\ \left(x^2 - y^2\right)\frac{1}{r^2} \\
Z^{\mathrm{s}}_{2,2}(\hat{\mathbf{e}}_r) &= \frac{1}{2}\sqrt{\frac{15}{\pi}}\ \frac{xy}{r^2} \\
Z^{\mathrm{c}}_{2,1}(\hat{\mathbf{e}}_r) &= \frac{1}{2}\sqrt{\frac{15}{\pi}}\ \frac{zx}{r^2} \\
Z^{\mathrm{s}}_{2,1}(\hat{\mathbf{e}}_r) &= \frac{1}{2}\sqrt{\frac{15}{\pi}}\ \frac{zy}{r^2} \\
Z_{2,0}(\hat{\mathbf{e}}_r) &= \frac{1}{4}\sqrt{\frac{5}{\pi}}\ \left(3z^2 - r^2\right)\frac{1}{r^2} \\
Z^{\mathrm{c}}_{3,3}(\hat{\mathbf{e}}_r) &= \frac{1}{4}\sqrt{\frac{35}{2\pi}}\ \left(x^3 - 3xy^2\right)\frac{1}{r^3} \\
Z^{\mathrm{s}}_{3,3}(\hat{\mathbf{e}}_r) &= \frac{1}{4}\sqrt{\frac{35}{2\pi}}\ \left(3x^2y - y^3\right)\frac{1}{r^3} \\
Z^{\mathrm{c}}_{3,2}(\hat{\mathbf{e}}_r) &= \frac{1}{4}\sqrt{\frac{105}{\pi}}\ \left(x^2 - y^2\right)\frac{z}{r^3} \\
Z^{\mathrm{s}}_{3,2}(\hat{\mathbf{e}}_r) &= \frac{1}{2}\sqrt{\frac{105}{\pi}}\ \frac{xyz}{r^3} \\
Z^{\mathrm{c}}_{3,1}(\hat{\mathbf{e}}_r) &= \frac{1}{4}\sqrt{\frac{21}{2\pi}}\ \left(5z^2 - r^2\right)\frac{x}{r^3} \\
Z^{\mathrm{s}}_{3,1}(\hat{\mathbf{e}}_r) &= \frac{1}{4}\sqrt{\frac{21}{2\pi}}\ \left(5z^2 - r^2\right)\frac{y}{r^3} \\
Z_{3,0}(\hat{\mathbf{e}}_r) &= \frac{1}{4}\sqrt{\frac{7}{\pi}}\ \left(5z^2 - 3r^2\right)\frac{z}{r^3}
\end{aligned}
$$

and so forth. These lead to the Cartesian-friendly, real-valued versions of the hydrogenic wavefunctions:

orbital	$\langle \mathbf{r} \vert \psi \rangle$
$1s$	$\sqrt{\frac{Z^3}{\pi a_0^3}}\ e^{-Zr/a_0}$
$2s$	$\frac{1}{4}\sqrt{\frac{Z^5}{2\pi a_0^5}}\ \left(r - 2\frac{a_0}{Z}\right)\ e^{-Zr/(2a_0)}$
$2p_x$	$\frac{1}{4}\sqrt{\frac{Z^5}{2\pi a_0^5}}\ x\ e^{-Zr/(2a_0)}$
$2p_y$	$\frac{1}{4}\sqrt{\frac{Z^5}{2\pi a_0^5}}\ y\ e^{-Zr/(2a_0)}$
$2p_z$	$\frac{1}{4}\sqrt{\frac{Z^5}{2\pi a_0^5}}\ z\ e^{-Zr/(2a_0)}$
$3s$	$\frac{2}{81}\sqrt{\frac{Z^7}{3\pi a_0^7}}\left(r^2 - 9\frac{a_0}{Z}r + \frac{27}{2}\left(\frac{a_0}{Z}\right)^2\right) e^{-Zr/(3a_0)}$
$3p_x$	$\frac{1}{81}\sqrt{\frac{2Z^7}{\pi a_0^7}}\ \left(r - 6\frac{a_0}{Z}\right)\ x\ e^{-Zr/(3a_0)}$
$3p_y$	$\frac{1}{81}\sqrt{\frac{2Z^7}{\pi a_0^7}}\ \left(r - 6\frac{a_0}{Z}\right)\ y\ e^{-Zr/(3a_0)}$
$3p_z$	$\frac{1}{81}\sqrt{\frac{2Z^7}{\pi a_0^7}}\ \left(r - 6\frac{a_0}{Z}\right)\ z\ e^{-Zr/(3a_0)}$
$3d_{3z^2-r^2}$	$\frac{1}{81}\sqrt{\frac{Z^7}{6\pi a_0^7}}\ (3z^2 - r^2)\ e^{-Zr/(3a_0)}$
$3d_{zx}$	$\frac{1}{81}\sqrt{\frac{2Z^7}{\pi a_0^7}}\ zx\ e^{-Zr/(3a_0)}$
$3d_{zy}$	$\frac{1}{81}\sqrt{\frac{2Z^7}{\pi a_0^7}}\ zy\ e^{-Zr/(3a_0)}$
$3d_{xy}$	$\frac{1}{81}\sqrt{\frac{2Z^7}{\pi a_0^7}}\ xy\ e^{-Zr/(3a_0)}$
$3d_{x^2-y^2}$	$\frac{1}{81}\sqrt{\frac{Z^7}{2\pi a_0^7}}\ (x^2 - y^2)\ e^{-Zr/(3a_0)}$

orbital	$\langle \mathbf{r} \vert \psi \rangle$
$4s$	$\frac{1}{1536}\sqrt{\frac{Z^9}{\pi a_0^9}}\,\left(r^3 - 24\frac{a_0}{Z}r^2 + 144\left(\frac{a_0}{Z}\right)^2 r - 192\left(\frac{a_0}{Z}\right)^3\right)\,e^{-Zr/(4a_0)}$
$4p_x$	$\mp\frac{1}{512}\sqrt{\frac{Z^9}{5\pi a_0^9}}\,\left(r^2 - 20\frac{a_0}{Z}r + 80\left(\frac{a_0}{Z}\right)^2\right)\,x\,e^{-Zr/(4a_0)}$
$4p_y$	$\mp\frac{1}{512}\sqrt{\frac{Z^9}{5\pi a_0^9}}\,\left(r^2 - 20\frac{a_0}{Z}r + 80\left(\frac{a_0}{Z}\right)^2\right)\,y\,e^{-Zr/(4a_0)}$
$4p_y$	$\mp\frac{1}{512}\sqrt{\frac{Z^9}{5\pi a_0^9}}\,\left(r^2 - 20\frac{a_0}{Z}r + 80\left(\frac{a_0}{Z}\right)^2\right)\,z\,e^{-Zr/(4a_0)}$
$4d_{3z^2-r^2}$	$\frac{1}{3072}\sqrt{\frac{Z^9}{\pi a_0^9}}\,\left(r - 12\frac{a_0}{Z}\right)\,\left(3z^2 - r^2\right)\,e^{-Zr/(4a_0)}$
$4d_{zx}$	$\frac{1}{512}\sqrt{\frac{Z^9}{3\pi a_0^9}}\,\left(r - 12\frac{a_0}{Z}\right)\,zx\,e^{-Zr/(4a_0)}$
$4d_{zy}$	$\frac{1}{512}\sqrt{\frac{Z^9}{3\pi a_0^9}}\,\left(r - 12\frac{a_0}{Z}\right)\,zy\,e^{-Zr/(4a_0)}$
$4d_{xy}$	$\frac{1}{512}\sqrt{\frac{Z^9}{3\pi a_0^9}}\,\left(r - 12\frac{a_0}{Z}\right)\,xy\,e^{-Zr/(4a_0)}$
$4d_{x^2-y^2}$	$\frac{1}{1024}\sqrt{\frac{Z^9}{3\pi a_0^9}}\,\left(r - 12\frac{a_0}{Z}\right)\,\left(x^2 - y^2\right)\,e^{-Zr/(4a_0)}$
$4f_{xyz}$	$\frac{1}{512}\sqrt{\frac{Z^9}{\pi a_0^9}}\,xyz\,e^{-Zr/(4a_0)}$
$4f_{z(x^2-y^2)}$	$\frac{1}{1024}\sqrt{\frac{Z^9}{\pi a_0^9}}\,z\left(x^2 - y^2\right)\,e^{-Zr/(4a_0)}$
$4f_{y(3x^2-y^2)}$	$\frac{1}{1024}\sqrt{\frac{Z^9}{6\pi a_0^9}}\,y\left(3x^2 - y^2\right)\,e^{-Zr/(4a_0)}$
$4f_{x(x^2-3y^2)}$	$\frac{1}{1024}\sqrt{\frac{Z^9}{6\pi a_0^9}}\,x\left(x^2 - 3y^2\right)\,e^{-Zr/(4a_0)}$
$4f_{x(4z^2-x^2-y^2)}$	$\frac{1}{1024}\sqrt{\frac{Z^9}{10\pi a_0^9}}\,x\left(4z^2 - x^2 - y^2\right)\,e^{-Zr/(4a_0)}$
$4f_{y(4z^2-x^2-y^2)}$	$\frac{1}{1024}\sqrt{\frac{Z^9}{10\pi a_0^9}}\,y\left(4z^2 - x^2 - y^2\right)\,e^{-Zr/(4a_0)}$
$4f_{z(2z^2-3x^2-3y^2)}$	$\frac{1}{1024}\sqrt{\frac{Z^9}{15\pi a_0^9}}\,z\left(2z^2 - 3x^2 - 3y^2\right)\,e^{-Zr/(4a_0)}$